



Neuro-Workflow: Agent-Assisted Brain Modeling



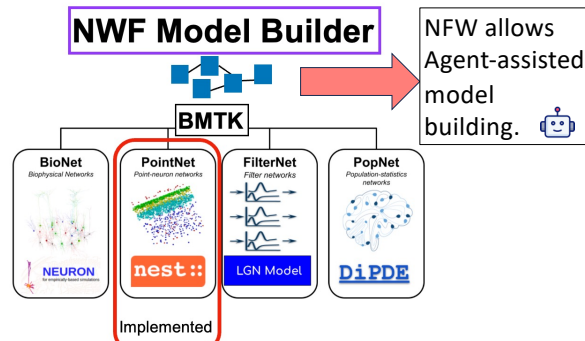
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The Brain/MINDS 2.0 Neuro-Workflow (NWF)

Neuro-Workflow is a node-based framework that builds and represents brain models as self-describing graphs, making it easy for researchers, non-experts, and AI agents to assemble, reuse, and extend them. A node is a Python class wrapping a modeling step and can integrate different tools and libraries.

Model Builder

We developed generic NWF nodes on top of BMTK (Allen Institute) for population, connectivity, stimulus, simulation configuration and analysis.



Interacting Populations in NEST

Our AI agent builds the workflow automatically

1 AI Assistant: ok, lets make a network...

ok, lets make a network of interacting populations.

Great—this project is currently empty. What kind of interacting-population model would you like to build?

For example, I can set up:

- two reciprocally coupled excitatory/inhibitory populations,
- a multi-region rate model with a connectivity matrix,
- a spiking NEST network driven by Poisson input, or a TVB whole-brain population model.

Tell me the simulator/model family and the number of populations or brain regions you have in mind.

NEST, exc and inh populations. Lets use the builder nodes NW_ nodes from neuro-workflow

Built a NEST/PointNet excitatory-inhibitory network using the 'NW_' builder nodes:

- Excitatory population: 800 'iaf_psc_alpha' neurons
- Inhibitory population: 200 'iaf_psc_alpha' neurons
- Sparse 10% random recurrent projections: E→E, E→I, I→E, and I→I
- Excitatory synapses: +5 pA; inhibitory synapses: -20 pA
- Synaptic delays: 1.5 ms for excitation and 1.0 ms for inhibition
- Added NEST simulation configuration and spike/membrane-trace analysis.

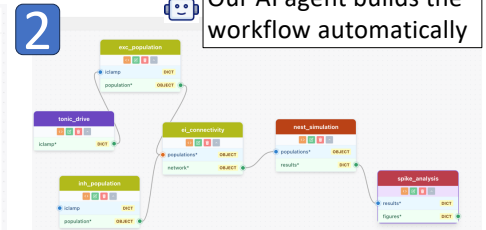
The populations are correctly passed through the fan-in connectivity builder, then to the NEST simulation and analysis stages.

I assembled a minimal two-population spiking E-I circuit using the NeuroWorkflow 'NW_' BMTK/NEST nodes.

Two 'NW_Population' components define point-neuron populations using the NEST 'iaf_psc_alpha' model:

- **Excitatory population ('exc')**: 800 neurons, labeled as excitatory.
- **Inhibitory population ('inh')**: 200 neurons, labeled as inhibitory.

Type a message...



Code generation and simulation

3

Neuro-Workflow v1.1.0

The model is represented in Sonata format

Basal Ganglia (NEST) using custom-made nodes

Model of the macaque BG (Gutierrez et al. 2025). Custom nodes are developed to cover a wider spectrum of models.

1 AI Assistant: Hi, lets do a simulation...

Hi, lets do a simulation. First I would like to search for a project, the macaque basal ganglia, and reproduce it in this workspace.

I found the reference project ***BG macaque*** a NEST 3 resting-state basal ganglia model for macaque. It contains four components for parameterization, network construction, simulation, and result writing (9 connections total).

I attempted to reproduce it in the current workspace, but the workspace update was rejected by the backend. I can retry by rebuilding the model component-by-component if you'd like.

yes, please rebuild the project here

Rebuilt the macaque basal ganglia resting-state simulation in this workspace.

It includes the parameter, NEST network construction, simulation, and results-writing components, with all 9 original scientific data connections restored. I also matched the source configuration: 1500 ms simulation duration, 300 ms warm-up, scale factor 0.7, MSN current 26 pA, and results written to './results'.

It is ready for code generation and execution.

thanks, now, what is the scale of the model? how many neurons?

The current model uses a **scale factor of 0.7**.

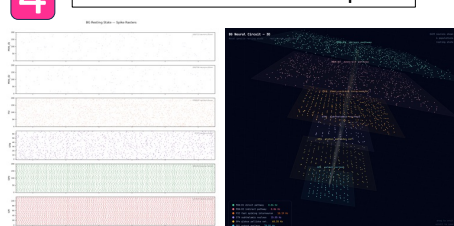
This means every basal-ganglia population is instantiated at **70% of the reference model's neuron count**. The reference scale factor of **1.0** corresponds to a published ***1834 downsampled macaque model***—approximately one simulated neuron per 834 biological neurons.

The exact total number of simulated neurons is determined internally from the population-specific anatomical counts and then multiplied by

Type a message...

2

Plots and interactive outputs



Marmoset Brain (TVB) using custom-made nodes

Whole-brain neural dynamics of the common marmoset were simulated based on a structural connectome derived from diffusion MRI tractography from Brain/MINDS 2.0 (BMCR data, Skibbe et al., 2023).

1 Generated workflow

TVBBrainViewerNode

Simulation Viewer

2

3 Our AI agent allows interaction with the simulation

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Repository: <https://github.com/oist/neuro-workflow>